

PROVISIONAL PATENT APPLICATION · 35 U.S.C. § 111(B)

SYSTEM AND METHOD FOR HIERARCHICAL DEVELOPMENTAL LANGUAGE ACQUISITION, MULTILINGUAL TRANSFER LEARNING, AND INTEGRATED SPEECH OUTPUT IN A FEDERATED QUERY SYSTEM

Applicant / Assignee: QueryBook Core Technology Group

Inventors: [to be inserted by counsel]

Related applications: Prior provisional applications on the QueryBook system, including the Universal Federated Connectivity System (UFCS) and the Federated Query Language (FQL).

Document date: September 28, 2026 (rev. September 29, 2026 — added Section VII reduced-to-practice embodiment; rev. October 1, 2026 — added Section VIII advanced deterministic multilingual pronunciation, self-contained speech, and quality-control embodiment) · **Attorney docket / Application no.:** [to be assigned]

Note on this draft. This provisional specification consolidates the Language Expression Layer (LEL) design of record. Clarifying refinements from an internal engineering audit — marked **AUDIT** — have been folded into the detailed description to strengthen enablement (operational threshold definitions, hybrid co-grounding, licensed training data, empirically-set timelines, non-strict multilingual acceleration). Formal drawings (FIG. 1–FIG. 8) with a Brief Description of the Drawings and a Reference Numerals key have been added and are draft — to be finalized to USPTO drawing standards (37 CFR 1.84) by a patent draftsman. Claims are reproduced as drafted and must be reviewed, and a prior-art search conducted, by qualified patent counsel before filing.

DRAWINGS

Formal patent drawings — black line art. Each figure is on its own sheet and every element is identified by a reference numeral used consistently in the Brief Description of the Drawings, the Reference Numerals key, and the Detailed Description. Figures are draft; counsel/draftsperson to finalize to USPTO drawing standards (37 CFR 1.84) before filing.

FIG. 1

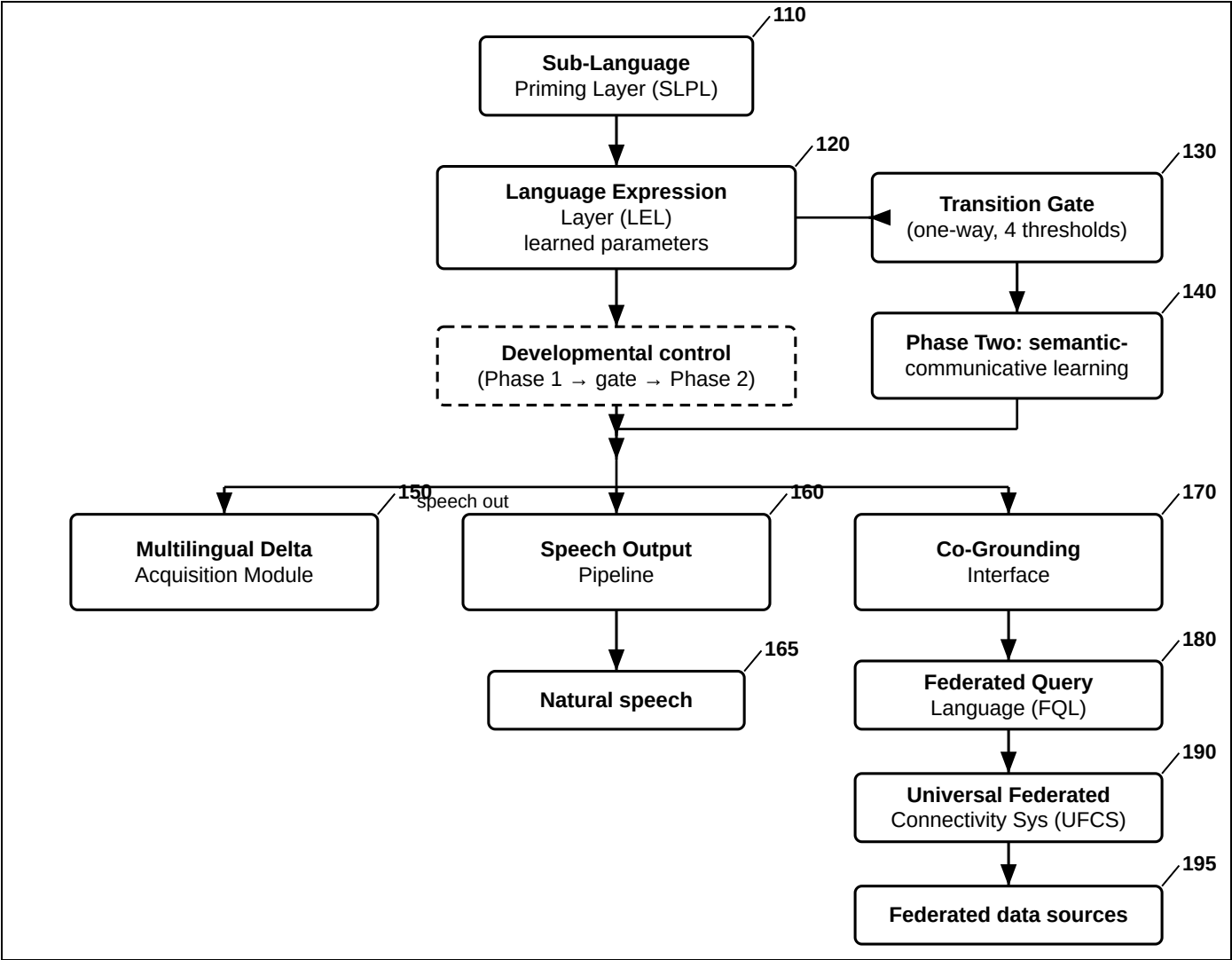


FIG. 1 — QueryBook LEL — Overall System Architecture

FIG. 2

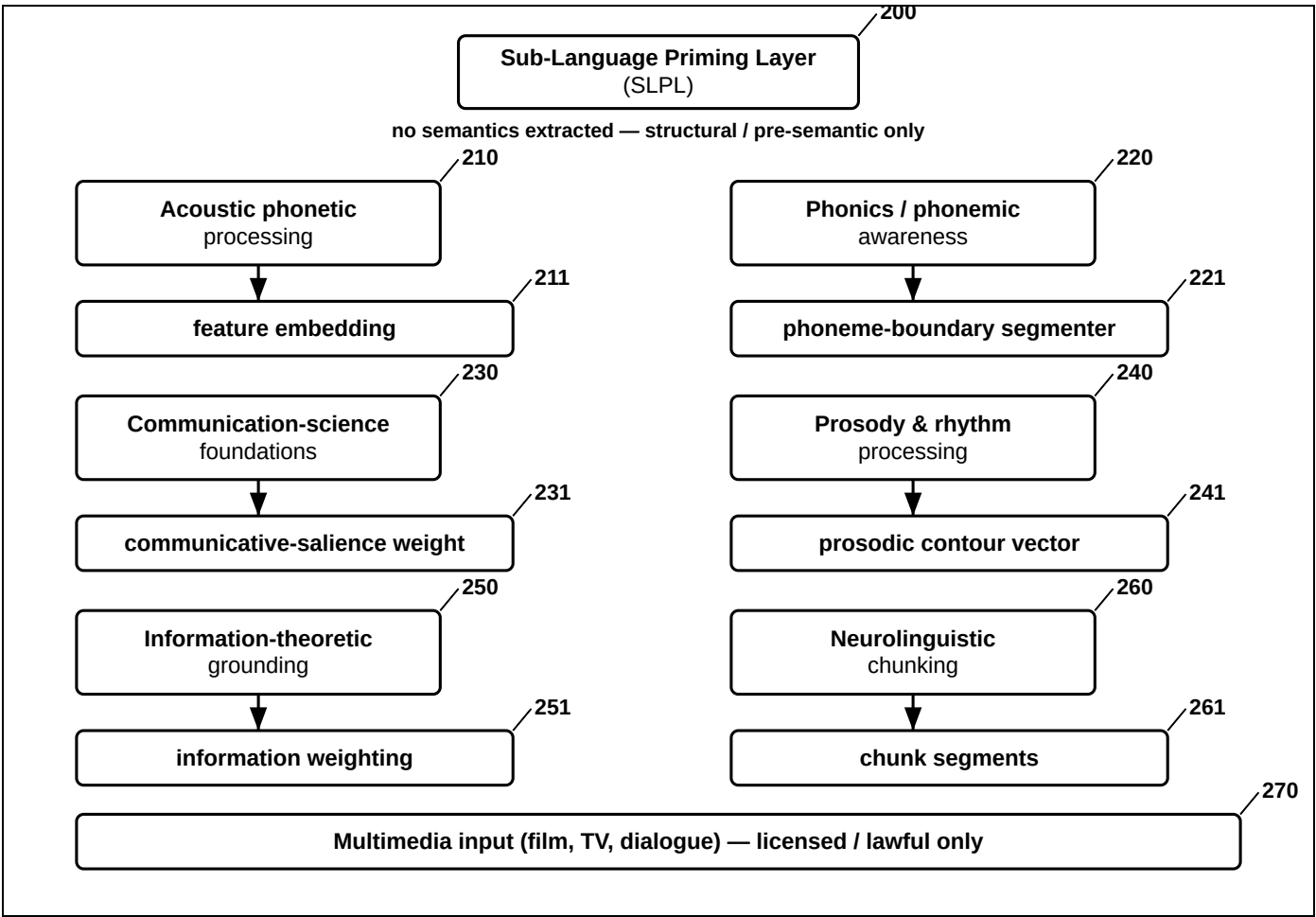


FIG. 2 — Sub-Language Priming Layer (SLPL) and Its Measurable Outputs

FIG. 3

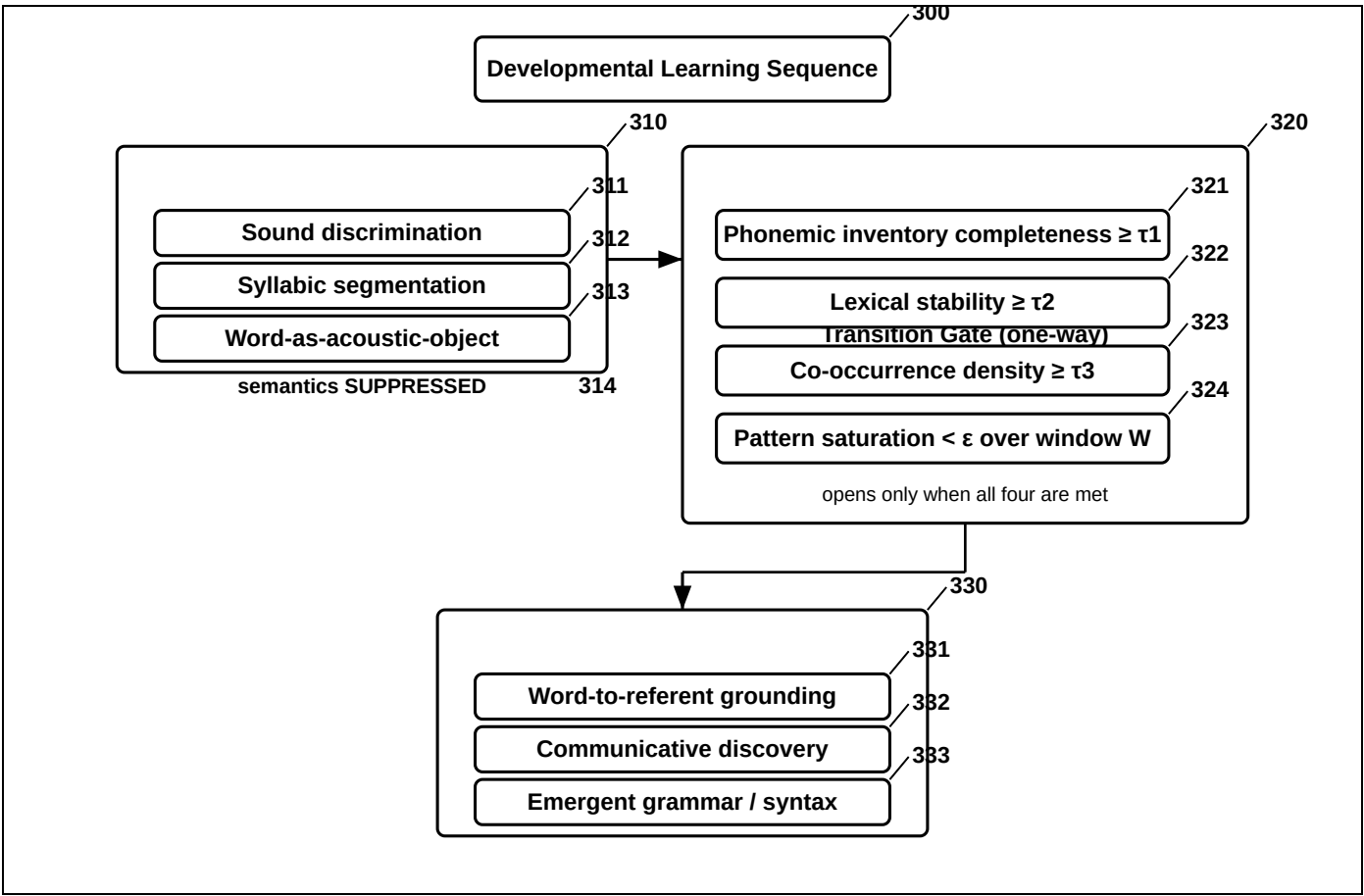


FIG. 3 — Developmental Learning Sequence and One-Way Transition Gate

FIG. 4

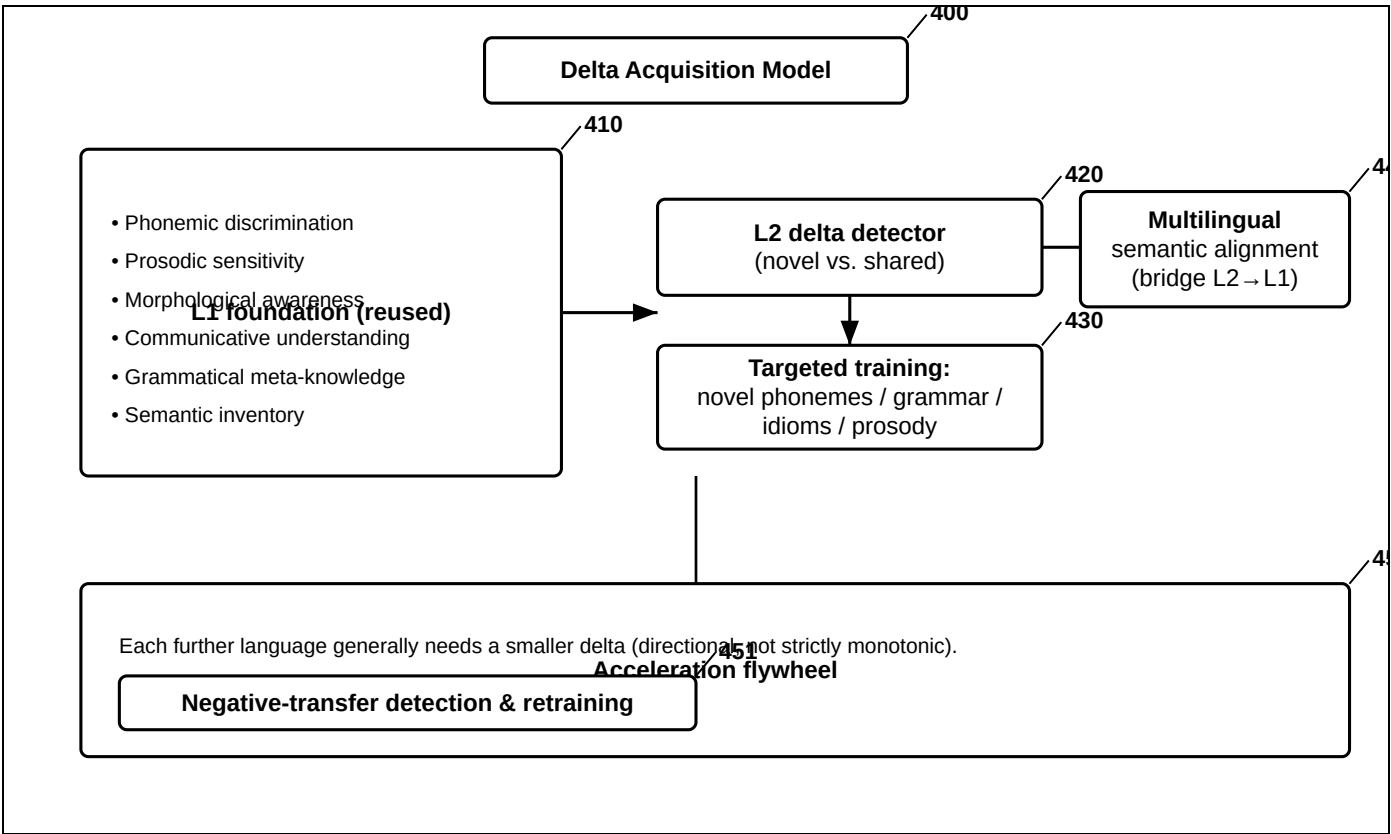


FIG. 4 — Multilingual Transfer — Delta Acquisition and Acceleration Flywheel

FIG. 5

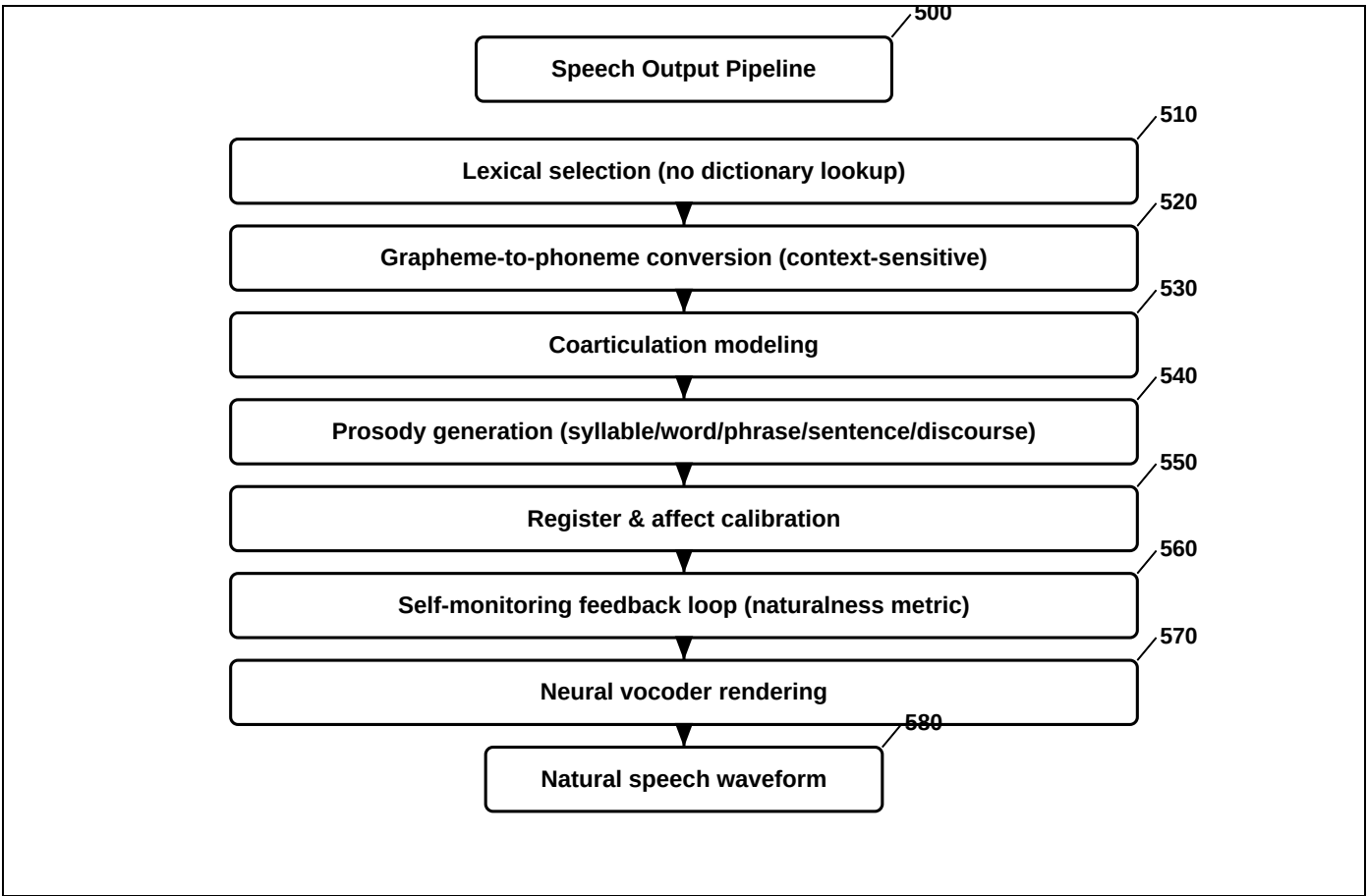


FIG. 5 — Speech Output Pipeline (Seven Learned Stages)

FIG. 6

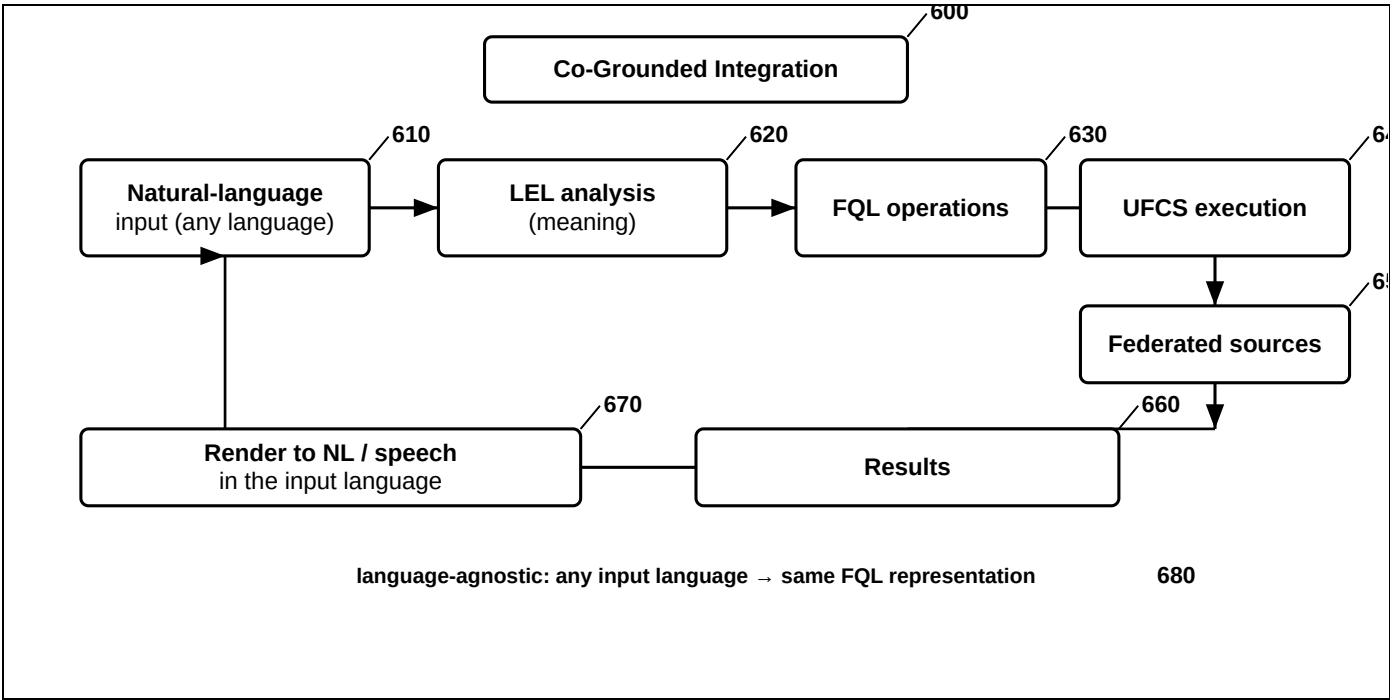


FIG. 6 — Co-Grounded Integration with FQL and UFCS

FIG. 7

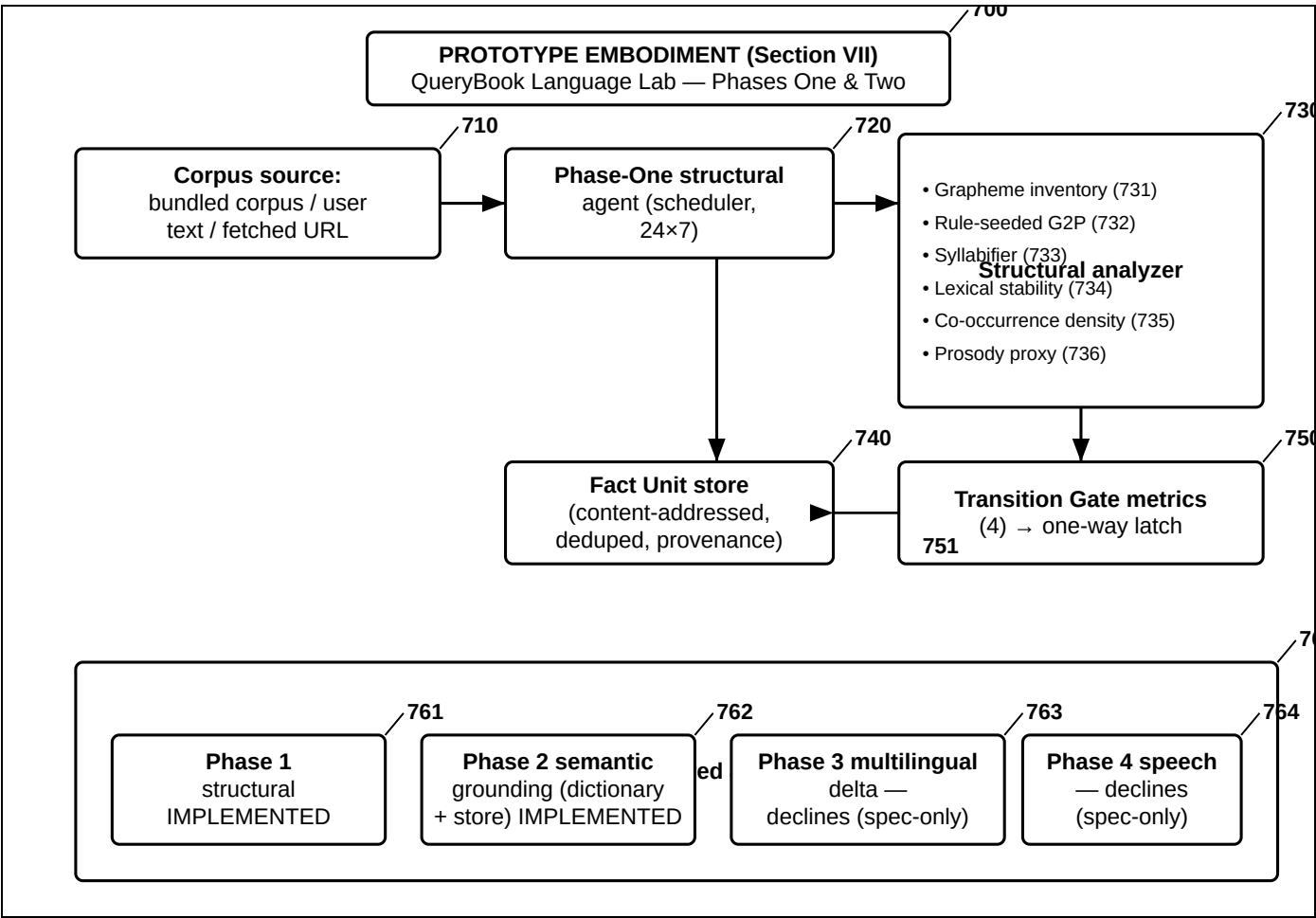


FIG. 7 — Reduced-to-Practice Embodiment — Language Lab Phases One and Two Prototype

FIG. 8

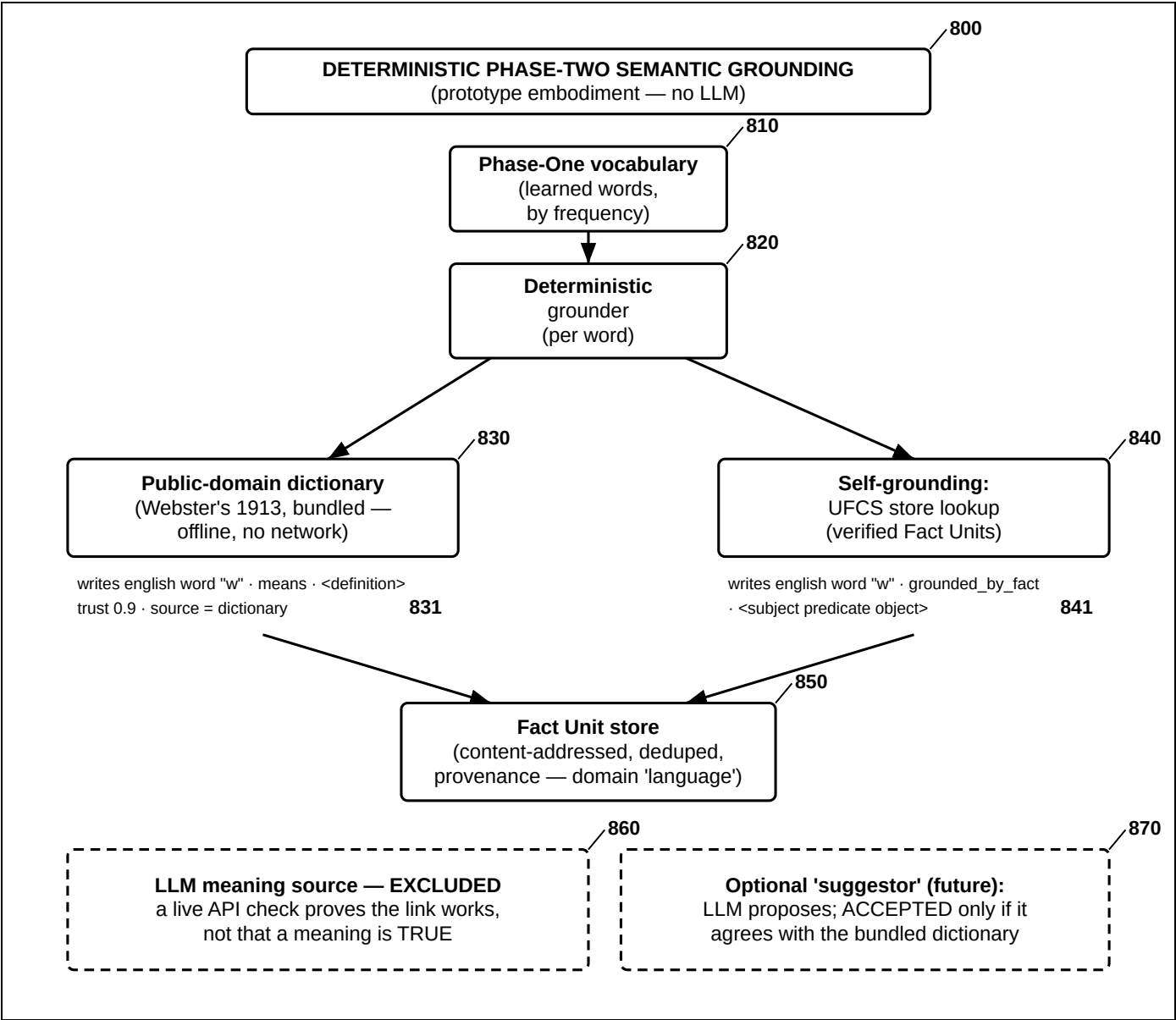


FIG. 8 — Deterministic Phase-Two Semantic Grounding — Dictionary and Store (No LLM)

FIG. 9

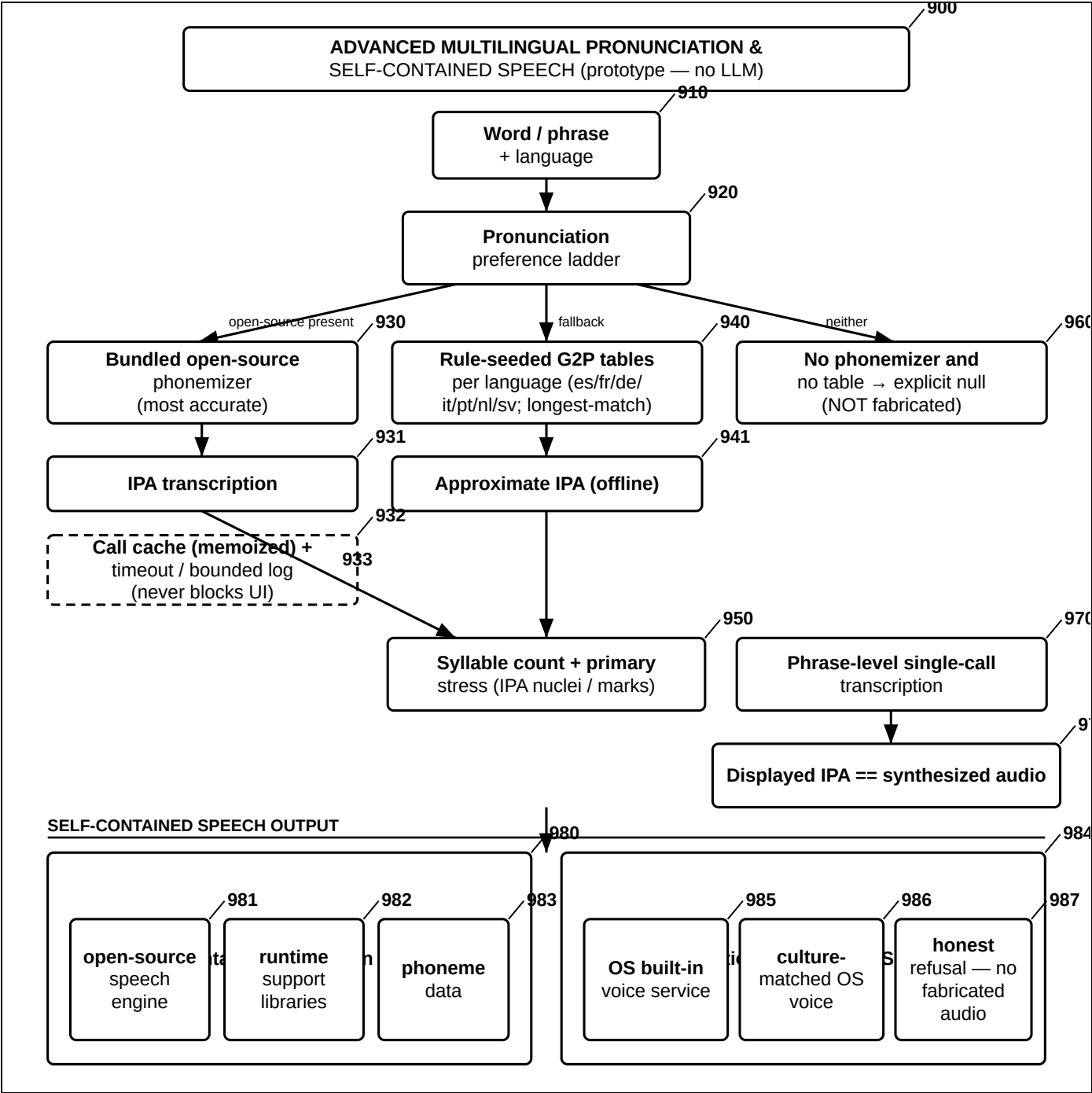


FIG. 9 — Advanced Multilingual Pronunciation and Self-Contained Speech (No LLM)

FIG. 10

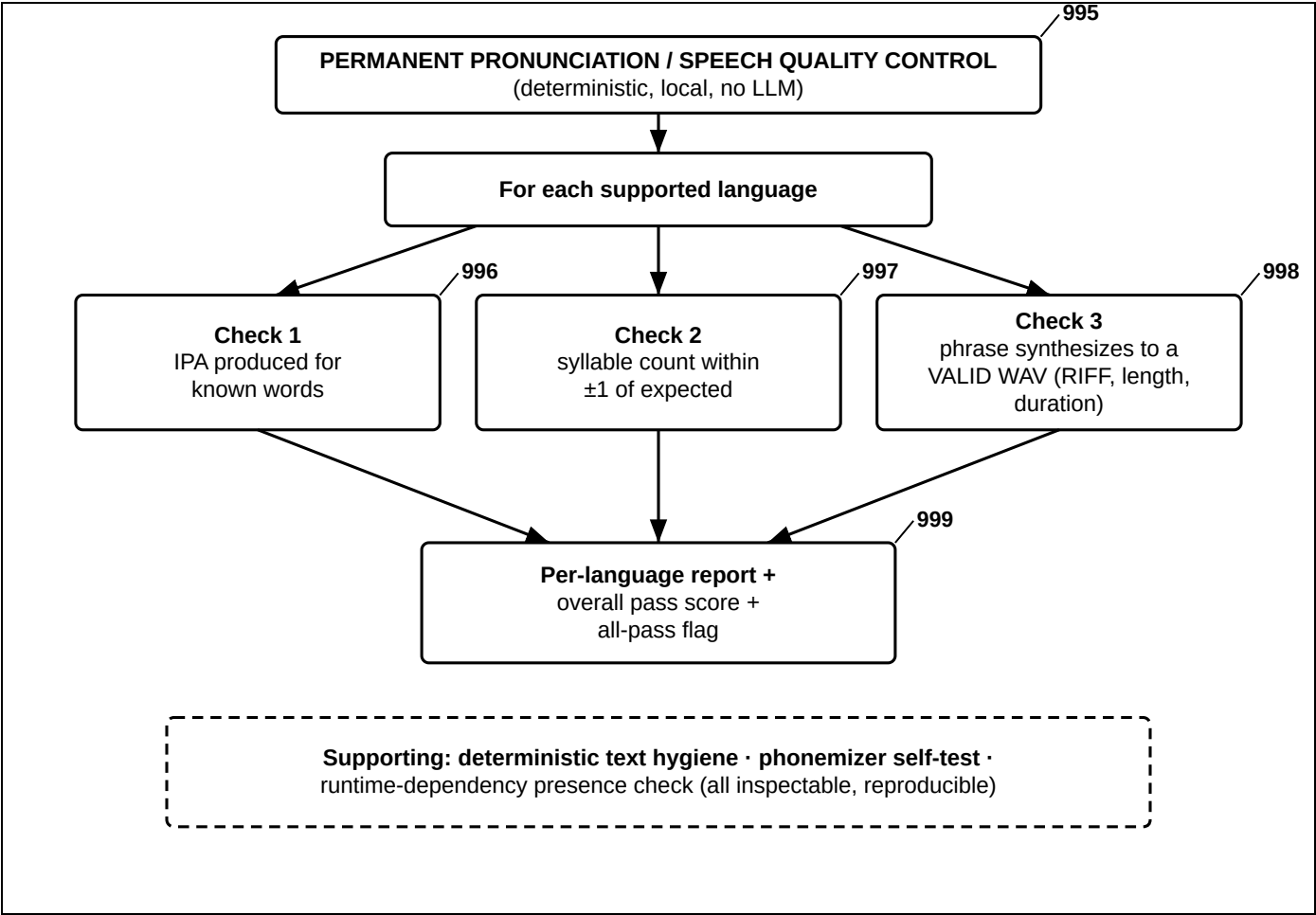


FIG. 10 — Permanent Pronunciation and Speech Quality-Control Battery (No LLM)

FIELD OF THE INVENTION

The present invention relates to machine learning systems for natural language acquisition and speech production. More particularly, it relates to a hierarchical, developmentally-sequenced language learning architecture — the Language Expression Layer (LEL) — that acquires linguistic capabilities from a structured training corpus without reliance on external dictionaries, rule-based modules, or language packs; a sub-language priming layer establishing perceptual and structural prerequisites before language learning begins; a transition gate governing progression from structural to semantic learning; an accelerated multilingual acquisition model exploiting first-language structural transfer; a multi-stage speech output pipeline producing natural speech from internalized learned parameters; and the co-grounded integration of the LEL with a Federated Query Language (FQL) and Universal Federated Connectivity System (UFCS).

BACKGROUND OF THE INVENTION

[0001] Conventional natural language processing systems acquire linguistic capability through (1) rule-based systems encoding grammar, morphology, syntax, and pronunciation via manually authored rules, dictionaries, and language packs; or (2) statistical and neural language models trained top-down on tokenized text, deriving associations between tokens without grounding in sub-linguistic or perceptual foundations.

[0002] Both paradigms have fundamental limitations. Rule-based systems are brittle, language-specific, and do not generalize beyond their rules. Neural models trained top-down on text lack grounding in the acoustic, prosodic, and communicative

substrates from which human language emerges. Neither replicates the developmental sequence of human language acquisition.

[0003] Human language acquisition proceeds developmentally: an infant performs sub-linguistic perceptual processing — discriminating acoustic signals, detecting prosody, segmenting continuous sound — before apprehending semantics. Words are first acquired as stable acoustic objects. The understanding that words serve communication emerges as a discovered insight, not a pre-loaded assumption. Grammar and syntax develop subsequently as tools serving communication.

[0004] Conventional systems also treat multilingual capability as parallel resource construction — separate dictionaries, grammars, and pronunciation databases per language — failing to exploit the structural transfer by which knowledge of a first language accelerates acquisition of later languages.

[0005] Existing speech synthesis relies on externally authored pronunciation lexicons, rule-based phoneme-to-grapheme mappings, and separately trained prosody and vocoder modules — engineered rather than learned.

[0006] There exists a need for a language acquisition and speech production system that: (a) replicates the human developmental sequence from sub-linguistic foundations to emergent communicative and semantic competence; (b) internalizes all linguistic features as learned parameters within a unified model, without external dictionaries, rule engines, or language packs; (c) exploits first-language structural transfer to accelerate multilingual acquisition; (d) produces natural speech through an integrated learned pipeline; and (e) co-grounds linguistic understanding with structured federated data query operations.

SUMMARY OF THE INVENTION

The invention provides a system and method for hierarchical developmental language acquisition and integrated speech output in a federated query system, comprising: (1) a Sub-Language Priming Layer establishing acoustic, phonemic, prosodic, communicative, and information-theoretic foundations before language learning; (2) a Language Expression Layer (LEL), a trained model acquiring all linguistic capabilities as learned parameters from a structured multilingual corpus; (3) a Transition Gate of four measurable readiness thresholds governing progression from structural-acoustic to semantic-communicative learning; (4) a Multilingual Transfer (Delta Acquisition) model reusing first-language foundations to accelerate later languages; (5) a Speech Output Pipeline of sequentially integrated learned stages; and (6) a Co-Grounded Integration architecture linking the LEL to FQL and UFCS for language-agnostic query execution.

No large language model acquires language, grounds meaning, or asserts a fact. Unlike conventional systems, the invention acquires and performs language understanding without any large language model (LLM). The LEL learns its own parameters bottom-up from the Sub-Language Priming Layer rather than being trained on, distilled from, or querying a pre-existing LLM; and in the reduced-to-practice embodiment (Section VII) semantic grounding is performed deterministically, drawing word meanings only from a bundled public-domain dictionary and by self-grounding against verified Fact Units in the store, each meaning carrying an auditable source. An LLM is expressly excluded from the language-understanding and grounding path (see FIG. 8), and no fact asserted by the system originates from an LLM. An LLM **may**, however, be used elsewhere in strictly non-asserting roles: (i) to phrase or narrate an answer over facts the system has already verified, and (ii) to propose hypotheses and drive simulations for exploration — in each case the LLM never establishes truth, and any candidate it proposes is admitted only after independent verification against the deterministic store (an optional grounding “suggestor” is admitted only where it agrees with the public-domain dictionary, which remains the authority).

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

FIG. 2 is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

FIG. 3 is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

FIG. 4 is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

FIG. 5 is a block diagram of the seven-stage Speech Output Pipeline.

FIG. 6 is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

FIG. 7 is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two, per Section VII.

FIG. 8 is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

FIG. 9 is a block diagram of the advanced multilingual pronunciation subsystem and self-contained speech output of the reduced-to-practice embodiment, showing the pronunciation preference ladder (bundled open-source phonemizer, then per-language rule-seeded grapheme-to-phoneme tables, then an explicit non-fabricated null), the single-call phrase transcription that keeps the displayed transcription consistent with the synthesized audio, and the self-contained bundle and engine-selection ladder that produce offline audio or an honest refusal without fabricating audio, per Section VIII.

FIG. 10 is a flow diagram of the permanent, deterministic pronunciation-and-speech quality-control battery of the reduced-to-practice embodiment, which, for each supported language and without a large language model, verifies that a phonemic transcription is produced, that the syllable count is within tolerance, and that a short phrase synthesizes to a valid audio waveform, returning a per-language report and an overall pass score, per Section VIII.

Reference Numerals

No.	Element	No.	Element
100	QueryBook LEL system	650	federated sources
110	Sub-Language Priming Layer (SLPL)	660	results
120	Language Expression Layer (LEL)	670	render to NL / speech in input language
130	Transition Gate	680	language-agnostic resolution
140	Phase Two semantic-communicative learning	700	Language Lab prototype embodiment
150	Multilingual Delta Acquisition module	710	corpus source
160	Speech Output Pipeline	720	Phase-One structural agent
165	natural speech output	730	structural analyzer
170	co-grounding interface	731	grapheme inventory
180	Federated Query Language (FQL)	732	rule-seeded G2P
190	Universal Federated Connectivity System (UFCS)	733	syllabifier
195	federated data sources	734	lexical-stability metric
200	SLPL	735	co-occurrence density metric
210	acoustic phonetic processing	736	prosody proxy

No.	Element	No.	Element
211	signal-level feature embedding	740	Fact Unit store
220	phonics / phonemic awareness	750	Transition-Gate metrics
221	phoneme-boundary segmenter	751	one-way latch
230	communication-science foundations	760	phase agents on shared manager
231	communicative-salience weight	761	Phase-1 agent (implemented)
240	prosody & rhythm processing	762	Phase-2 agent — deterministic dictionary + store grounding (implemented)
241	prosodic contour vector	763	Phase-3 agent (declines — spec-only)
250	information-theoretic grounding	764	Phase-4 agent (declines — spec-only)
251	information weighting	800	deterministic Phase-Two semantic grounding
260	neurolinguistic chunking	810	Phase-One vocabulary (learned words)
261	chunk segments	820	deterministic grounder (per word)
270	licensed multimedia input	830	bundled public-domain dictionary source
300	developmental learning sequence	831	dictionary meaning Fact Unit (means, trust 0.9)
310	Phase One (structural-acoustic)	840	self-grounding store-lookup source
311	sound discrimination	841	store-link Fact Unit (grounded_by_fact)
312	syllabic segmentation	850	Fact Unit store (provenance, domain language)
313	word-as-acoustic-object	860	excluded large-language-model meaning source
314	semantics suppressed	870	optional future dictionary-gated suggestor
320	Transition Gate (one-way)	900	advanced multilingual pronunciation subsystem
321	phonemic-inventory completeness $\geq \tau_1$	910	pronunciation request (word/phrase, language)
322	lexical stability $\geq \tau_2$	920	pronunciation preference-ladder selector
323	co-occurrence density $\geq \tau_3$	930	bundled open-source phonemizer
324	pattern saturation $< \epsilon$	931	phonemizer IPA transcription
330	Phase Two (semantic-communicative)	932	phonemizer call cache
331	word-to-referent grounding	933	bounded call log / timeout guard
332	communicative discovery	940	rule-seeded G2P tables (per language)
333	emergent grammar / syntax	941	approximate rule-seeded IPA (offline)
400	Delta Acquisition Model	950	syllable count + primary-stress derivation
410	reused L1 foundation	960	explicit null result (not fabricated)
420	L2 delta detector	970	phrase-level single-call transcription
430	targeted training of novel elements	971	displayed transcription consistent with synthesized audio
440	multilingual semantic alignment	980	self-contained application bundle
450	acquisition acceleration flywheel	981	bundled open-source speech engine

No.	Element	No.	Element
451	negative-transfer detection & retraining	982	runtime-support libraries
500	Speech Output Pipeline	983	bundled phoneme data
510	lexical selection	984	engine-selection ladder (per OS / language)
520	grapheme-to-phoneme conversion	985	host OS built-in voice service
530	coarticulation modeling	986	culture-matched OS voice (when installed)
540	prosody generation	987	honest refusal with remedy (no fabricated audio)
550	register & affect calibration	990	deterministic translator (through English)
560	self-monitoring feedback loop	991	bundled open-source bilingual dictionary
570	neural vocoder rendering	992	coverage accounting + unknown-word passthrough (flagged)
580	natural speech waveform	993	deterministic language detector
600	co-grounded integration	995	permanent quality-control battery
610	natural-language input	996	IPA-produced check
620	LEL analysis	997	syllable-sanity check (± 1)
630	FQL operations	998	valid-waveform synthesis check
640	UFCS execution	999	per-language QC report + overall pass score

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Section I — The Sub-Language Priming Layer (SLPL) (FIG. 2)

[0007] The SLPL is a pre-training stage executing prior to language corpus ingestion, establishing the perceptual prerequisites that make language learnable. It comprises six components, each producing a defined, measurable representation **AUDIT**: (a) **Acoustic Phonetic Processing** — learns to discriminate sounds at the signal level (formant, pitch, duration, amplitude), producing a discrimination-thresholded feature embedding; (b) **Phonics and Phonemic Awareness** — learns sound-symbol correspondence, segmentation, blending, and minimal-pair discrimination, producing a discrete phoneme-boundary segmenter with a measured minimal-pair rate; (c) **Communication Science Foundations** — learns turn-taking and signal-versus-noise discrimination, producing a per-segment communicative-salience weight without assigning meaning; (d) **Prosody and Rhythm Processing** — learns stress, intonation, timing, and pause as pre-semantic carriers, producing a prosodic contour vector; (e) **Information-Theoretic Grounding** — acquires an entropy/redundancy weighting allocating representational attention by measured information density; and (f) **Neurolinguistic Chunking** — groups acoustic units into processable segments at a defined chunk-capacity window.

[0008] In a preferred embodiment, multimedia sources (motion pictures, television, dialogue recordings) are ingested during the SLPL phase, providing naturalistic prosody, rhythm, social turn-taking, and co-occurring audio-visual context. At this stage, content is processed for structural and prosodic properties only; no semantic content is extracted. **AUDIT** Only licensed or lawfully usable material is ingested for training.

Section II — The Language Expression Layer (LEL) (FIG. 1)

[0009] The LEL is a trained model acquiring all linguistic capabilities as internalized learned parameters, without reliance upon external dictionaries, pronunciation databases, rule-based grammar engines, morphological analyzers, or language packs at

inference. **AUDIT** Structured linguistic resources may be used as training inputs, but are internalized as parameters rather than retained as runtime lookup tables. The training corpus comprises multilingual text; phonetic data and transcriptions; pronunciation patterns; grammar structures; semantic alignment datasets; speech corpora; grapheme-to-phoneme mappings; phonology datasets; and multimedia corpora.

[0010] The LEL acquires, as learned parameters, lexical, morphological, grammatical/syntactic, semantic, idiomatic, phonetic/phonological, pronunciation, and prosodic knowledge. **AUDIT** The LEL is a multimodal (audio, text, video) training program whose duration is set by available compute and data and is established empirically. Corpus exposure is sequenced by the Transition Gate of Section III, which suppresses semantic alignment data during early training.

Section III — Developmental Learning Sequence and Transition Gate (FIG. 3)

[0011] **Phase One (Structural-Acoustic):** the LEL develops sound discrimination, syllabic segmentation, and word-as-acoustic-object recognition, with semantic alignment data suppressed as an explicit architectural constraint enforced by the Transition Gate.

[0012] The **Transition Gate** is a one-way measurable threshold mechanism. Phase Two commences only when all four thresholds are satisfied, each defined operationally with a calibratable target and a bounded no-convergence fallback **AUDIT** :

Threshold	Definition	Metric
Phonemic inventory completeness	Reliable discrimination of contrastive phonemes	minimal-pair accuracy $\geq \tau_1$
Lexical stability	Same word recognized across speaker/rate/register	cross-condition consistency $\geq \tau_2$
Co-occurrence density	Sufficient sound-context pairings for grounding	pairings/unit $\geq \tau_3$
Pattern saturation	Novel structural-pattern discovery has plateaued	novel-pattern rate $< \epsilon$ over window W

[0013] **Phase Two (Semantic-Communicative):** upon satisfaction of all thresholds, semantic alignment data is activated and the system performs word-to-referent grounding using co-occurrence of acoustic units with visual and contextual events; communicative discovery (learning that word production causes structured responses); and emergence of grammar, syntax, and semantic structure from distributional patterns rather than imposed rules.

Section IV — Multilingual Transfer and the Acquisition Flywheel (FIG. 4)

[0014] Following first-language (L1) acquisition, the system employs a **Delta Acquisition Model** for subsequent languages, learning only the genuinely novel elements while reusing L1 phonemic discrimination, prosodic sensitivity, morphological awareness, communicative understanding, grammatical meta-knowledge, and semantic inventory. A multilingual semantic alignment layer bridges L2 lexical items to existing L1 concepts.

[0015] Genuinely novel L2 elements — phonemes absent from L1, grammatical structures without L1 analogue, culturally grounded idioms, and differing prosodic systems — receive targeted training. **AUDIT** Acquisition cost generally decreases as shared structure accumulates (an acquisition flywheel); the trend is directional rather than strictly monotonic, and negative-transfer cases between particular language pairs are detected and resolved by targeted retraining.

Section V — The Speech Output Pipeline (FIG. 5)

[0016] The Speech Output Pipeline converts internalized representations into natural speech through seven sequential stages, each driven exclusively by learned parameters: (1) lexical selection from internalized lexical-semantic space without dictionary lookup; (2) context-sensitive grapheme-to-phoneme conversion, irregular forms handled by pattern matching; (3) coarticulation modeling; (4) prosody generation at syllable, word, phrase, sentence, and discourse levels; (5) register and affect calibration; (6)

a self-monitoring feedback loop evaluating the generated output against internalized acoustic targets and a defined naturalness metric **AUDIT** prior to emission; and (7) neural vocoder rendering, which renders only what upstream stages determine.

Section VI — Co-Grounded Integration with FQL and UFCS (FIG. 6)

[0017] The LEL is integrated with FQL and UFCS through a co-grounding relationship in which linguistic concepts and data entities share a common representational foundation. Natural-language input is analyzed by the LEL; the resulting semantic representations are mapped to FQL operations executed by the UFCS across federated data sources; results are rendered back to natural language, as text or speech, in the input language.

[0018] Because input in any supported language resolves to the same FQL representation, the query layer is language-agnostic.

AUDIT The correspondence between LEL semantics and the FQL schema is realized as a hybrid of an emergent structural correspondence, arising from the developmental grounding sequence, and, where necessary, an explicit semantic-alignment mapping layer between LEL concepts and FQL entities, relationships, and operations.

Section VII — Reduced-to-Practice Embodiment (Structural and Deterministic Semantic Phases) (FIG. 7, FIG. 8) **REV**

2026-09-30

[0019] A working embodiment of Phase One (structural-acoustic) of Section III has been reduced to practice as a deterministic, dependency-free software implementation (the “Language Lab”). The implementation ingests raw text and computes defined, measurable structural representations comprising: a grapheme inventory with frequencies; a rule-seeded grapheme-to-phoneme approximation providing an initial phoneme inventory; a syllabic segmentation; a lexical-stability measure; a word co-occurrence density; and prosodic proxies derived from punctuation and sentence structure. Each computed measurement is written as a provenance-tracked record (a Fact Unit) into a content-addressed, de-duplicated store, such that every structural observation carries a source, a confidence value, and a content-derived fingerprint.

[0020] The Transition Gate of Section III is embodied by evaluating four computed metrics — phonemic-inventory completeness, lexical stability, co-occurrence density, and pattern saturation — against calibratable thresholds, the gate latching one-way upon simultaneous satisfaction of all four. In the embodiment, structural learning is driven autonomously by a background agent that repeatedly ingests corpus material on a schedule, advancing the four metrics until the gate opens, at which point Phase One is complete. The embodiment demonstrates the developmental suppression of semantics during Phase One and the one-way gate mechanism.

[0021] In a further reduced-to-practice extension, Phase Two semantic grounding is embodied deterministically and **without any large language model** (FIG. 8). For words learned in Phase One, the embodiment attaches meaning from two auditable sources: (a) a bundled public-domain dictionary (for example, Webster’s Unabridged Dictionary of 1913), writing a provenance-tracked record of the form *english word “w” · means · <definition>* at a defined confidence with the dictionary as source; and (b) self-grounding, linking the word to verified Fact Units in which it already appears, of the form *english word “w” · grounded_by_fact · <subject predicate object>*. Each grounded meaning is a provenance-tracked Fact Unit carrying an auditable source; no large language model participates in the grounding path. An optional “suggestor” may propose a candidate meaning only where it agrees with the public-domain dictionary, which remains the authority.

[0022] The embodiment realizes the developmental phases as distinct agent roles upon a shared agent manager: a Phase-One structural agent and a Phase-Two deterministic semantic-grounding agent, both implemented and operative; and Phase-Three multilingual-delta and Phase-Four speech-output agents, which are specified and, pending their respective engines, decline to execute rather than produce unsupported output. **AUDIT** This prototype evidences enablement of the sub-language priming of Section I, the structural-acoustic phase and transition gate of Section III, the deterministic semantic grounding of Phase Two, and the co-grounding of structural observations with the federated store. The trained neural LEL of Section II, the Delta Acquisition Model of Section IV, and the Speech Output Pipeline of Section V are described as preferred embodiments and are

not asserted to be reduced to practice by this prototype; the deterministic dictionary-and-store semantic grounding of Phase Two **is** reduced to practice as described in FIG. 8, whereas the trained-neural form of Phase-Two grounding remains a preferred embodiment; the rule-seeded grapheme-to-phoneme mapping is an initial seeding inventory and not a learned grapheme-to-phoneme model; and language understanding and fact grounding throughout are achieved without any large language model. Where an LLM is employed at all — to phrase answers over already-verified facts, or to propose hypotheses and simulations — it operates in a non-asserting role and never establishes a fact the system has not independently verified against the deterministic store.

[0023] The embodiment further reduces to practice a family of reasoning agents on the shared agent manager, each observing the assertion boundary of the Prime Directive. A **hypothesis** agent and a **simulation** agent employ a large language model in a strictly non-asserting role — the former proposing candidate facts, the latter projecting the consequences of a scenario — whereupon each candidate is independently verified against the deterministic store (recorded as corroborated, contradicted, or open) and retained in isolation at a confidence below the assertion threshold, so that nothing proposed by the model is ever returned as a verified fact. A **gated reasoning** agent embodies the principle that deduction may assert while induction and abduction may only propose: it computes a deterministic transitive closure over verified premises — for a relation defined as transitive, from verified $X R Y$ and $Y R Z$ it asserts $X R Z$, each derived record carrying its two premises as provenance and a confidence that decays with each inferential hop so that chains terminate — requiring no large language model, because the conclusion is entailed by facts already verified; while inductive and abductive candidates are proposed and verified in the same non-asserting manner as the hypothesis agent. **AUDIT** An answer gate distinguishes multi-valued relations (for example *is_a*, *part_of*, *located_in*), for which a subject may hold several objects concurrently without contradiction, from functional relations, for which competing objects are reported as a contradiction. A **planner** agent decomposes a supplied goal into an ordered sequence of operational intents and **halts on ambiguity**, returning a clarifying question rather than fabricating a plan for an underspecified goal; the resulting plan is retained as a non-asserting proposal and asserts no facts.

[0024] The embodiment further reduces to practice a deterministic approximation of the multilingual Delta Acquisition Model of Section IV. Rather than a trained neural cross-lingual alignment, the embodiment reuses the English vocabulary acquired in Phase One and learns only the delta to a further language — the word-to-word mapping — from a bundled bilingual dictionary, recording each mapping as a provenance-tracked translation Fact Unit (*english word “w” · translation_<language> · <term>*) with an auditable source, and without any large language model. The bundled bilingual data is a demonstration set to be replaced with a public-domain or permissively-licensed source before commercial use; the trained neural cross-lingual alignment remains a preferred (roadmap) embodiment. Of the four developmental phases, Phases One through Three are thus reduced to practice deterministically; Phase Four is reduced to practice in its deterministic pronunciation-analysis and operating-system voice forms as described in [0025], while the trained neural speech-output pipeline of Section V remains a preferred embodiment.

[0025] In a further reduced-to-practice extension, the speech-output stage of Phase Four is embodied in two deterministic forms without any large language model. First, a pronunciation-analysis form computes, for each word learned in Phase One, a rule-seeded grapheme-to-phoneme sequence, a syllabic count, a heuristic primary-stress position, and a punctuation-derived prosodic contour, each written as a provenance-tracked Fact Unit (of the form *english word “w” · pronunciation · <phonemic transcription>*, *· syllable_count · <n>*, and *· stress_syllable · <k>*). Second, an audio-rendering form produces an actual speech waveform by invoking either the host operating system’s built-in text-to-speech facility or a bundled open-source speech engine, in each case through a subprocess interface and without reliance on any network service or account; where no such facility is present the embodiment reports its absence and **does not fabricate audio**. The particular speech engine is addressed only through this role interface and is substitutable, as further described in Section VIII. These deterministic forms evidence enablement of the speech-output stage; the trained neural speech-output pipeline of Section V and claims 10–13, comprising a learned coarticulation model, internalized multi-level prosody, and a neural vocoder, remains a preferred embodiment and is **not** asserted to be reduced to practice by this prototype.

[0026] The embodiment further contemplates, as a preferred (prophetic) embodiment building upon the reduced-to-practice hypothesis and simulation agents of [0023], a creativity-parity process directed to achieving language and creative-composition quality comparable or superior to that of a large language model while preserving the assertion boundary of the Prime Directive. In this process the hypothesis and simulation agents propose candidate compositions or answers, the model acting solely as a proposer; each candidate and the verdict of its verification against the deterministic store are recorded, timestamped, in an archive; the archive is analyzed over time for coverage, corroboration rate, novelty, and contradiction rate; and the system's output is compared against a version-pinned reference large language model on a fixed, versioned benchmark of grounded-answer, language-fluency, creative-composition, and multilingual tasks, scored both for quality and for compliance metrics comprising citation coverage, absence of fabrication, and deterministic reproducibility. Throughout, the model only proposes and never asserts; nothing it generates becomes an asserted fact absent independent verification against the deterministic store. This creativity-parity process, its archive, and its benchmark are described as a preferred embodiment and are **not** asserted to be reduced to practice by this prototype.

Section VIII — Reduced-to-Practice Embodiment (Advanced Multilingual Pronunciation, Self-Contained Speech, and Quality Control) (FIG. 9, FIG. 10) REV 2026-10-01

[0027] In a further reduced-to-practice extension of the pronunciation stage, the embodiment provides a **multilingual pronunciation subsystem** (900) that, for a supplied word or phrase in any supported language (910), derives a phonemic transcription, a syllable count, and a primary-stress position deterministically and without any network service, by a defined preference order. A selector (920) first determines whether a bundled **open-source phonemizer** (930) is present and, if so, obtains its transcription (931); phonemizer invocations are memoized in a cache (932) and bounded by a call log and timeout guard (933) so that no invocation blocks the interface. Where the phonemizer is absent, the subsystem applies a bundled **rule-seeded grapheme-to-phoneme table** specific to the language (940), comprising ordered longest-match rules, to produce an approximate transcription (941), thereby guaranteeing a transcription offline with no installed engine. Syllable count and stress (950) are derived from the vowel nuclei of the transcription together with the engine's stress marks or, for the rule form, a per-language stress rule. Where neither a phonemizer nor a rule table exists for the language, the subsystem returns an explicit **null** result (960) and **does not fabricate** a pronunciation.

[0028] In a further extension, a **phrase-level transcription form** (970) transcribes an entire phrase in a single phonemizer invocation rather than word by word, such that the phonetic transcription presented to the user is **identical to the transcription from which the audio waveform is synthesized** (971), eliminating divergence between a displayed transcription and the spoken output. For a first language processed by the rule form, or where no phonemizer is present, the per-word rule transcription of [0027] is used.

[0029] In a further extension directed to operation on commodity machines lacking any installed speech facility, the audio-rendering form of [0025] is embodied as a **self-contained application bundle** (980) that includes, within the application, an **open-source speech engine** (981), its **runtime-support libraries** (982), and its **phoneme data** (983), such that a machine with no previously installed voice produces real, language-correct audio offline, with no installation, account, or network. An **engine-selection ladder** (984) chooses, per operating system and per language, among the bundled open-source engine (981), the host operating system's built-in voice service (985), and a culture-matched operating-system voice where one is installed (986), and reports which engine produced each waveform; where no real voice exists for the language, the ladder returns an **honest refusal with a remedy** (987) and **does not fabricate audio**. The speech engine is addressed only through a role interface and is substitutable — for licensing, quality, or platform reasons — without altering the determinism or provenance guarantees of the system.

[0030] In a further extension of the multilingual delta of [0024], a **deterministic translator** (990) renders text between English and a supported language word-by-word through English from a bundled **open-source bilingual dictionary** (991), wherein a word absent from the dictionary is **passed through unchanged and flagged** rather than guessed, and a **coverage measure**

(992) reports the proportion of words rendered, so that the operation is fully auditable. A **language detector** (993) determines the language of supplied text deterministically by scoring the text against each bundled dictionary and reporting a ranked best estimate with a confidence value. No large language model participates in translation or detection.

[0031] In a further extension, a **permanent quality-control battery** (995) provides a deterministic, locally-executed gate on the pronunciation and speech subsystems that requires no large language model and may be executed at any time. For each supported language it verifies that a phonemic transcription is produced for known words (996); that the computed syllable count is within a bounded tolerance of an expected value (997); and that a short phrase **actually synthesizes to a valid audio waveform** — a well-formed container of non-trivial length and decodable duration (998); and it returns a per-language report together with an overall pass score and an all-pass indicator (999). **AUDIT** The battery, together with deterministic input hygiene, a phonemizer self-test, and a runtime-dependency presence check, evidences enablement of the pronunciation and speech subsystems and provides a standing, reproducible verification that they operate. These advanced deterministic forms are reduced to practice as described; the trained-neural pronunciation and speech-output forms of Sections II and V remain preferred embodiments.

CLAIMS

Claims are not required for a provisional application but are included to frame the inventive concepts; counsel to finalize on conversion to a non-provisional.

1. A computer-implemented system for hierarchical developmental language acquisition, comprising: a sub-language priming module configured to establish acoustic phonetic discrimination, phonemic awareness, prosodic sensitivity, communication-science foundations, information-theoretic signal weighting, and neurolinguistic chunking in a machine learning model prior to language corpus training; a Language Expression Layer (LEL) comprising a trained neural model configured to acquire, as internalized learned parameters, linguistic capabilities including phonetics, phonology, morphology, grammar, syntax, semantics, idiomatic expressions, and pronunciation across one or more languages from a training corpus, without reliance on external dictionaries, rule-based grammar modules, pronunciation databases, or language packs; and a transition gate mechanism configured to enforce a two-phase learning sequence in which a first phase of structural-acoustic learning completes and four readiness thresholds are satisfied before a second phase of semantic and communicative learning commences.
2. The system of claim 1, wherein the four thresholds comprise: a phonemic inventory completeness threshold measuring reliable discrimination of phonemically contrastive signals; a lexical stability threshold measuring consistent word-sound identification across varying acoustic conditions; a co-occurrence density threshold measuring accumulation of sound-context pairings sufficient for statistical grounding; and a pattern saturation threshold measuring diminishing novel structural-pattern discovery.
3. The system of claim 1, wherein the training corpus comprises multilingual text, phonetic data and transcriptions, pronunciation patterns, grammar structures, semantic alignment datasets, speech corpora, grapheme-to-phoneme mappings, phonology datasets, and multimedia sources comprising motion pictures, television, and dialogue recordings.
4. The system of claim 1, wherein semantic alignment data within the training corpus is suppressed during the first phase and activated only upon satisfaction of all four thresholds.
5. The system of claim 1, wherein during the first phase words are processed as stable acoustic objects without semantic association, and the second phase commences upon the system's derivation from the corpus that word units produce structured responses in social-interaction contexts.
6. A computer-implemented method for accelerated multilingual language acquisition comprising: acquiring a first language (L1) through a complete developmental sequence comprising sub-language priming, structural-acoustic learning, transition-gate

evaluation, and semantic-communicative learning; establishing from L1 a foundation comprising phonemic discrimination, prosodic sensitivity, morphological awareness, communicative understanding, grammatical meta-knowledge, and a semantic concept inventory; upon initiating a second language (L2), identifying a delta comprising phonemic, grammatical, morphological, and semantic elements present in L2 but absent from L1; and acquiring the delta while reusing the L1 foundation for transferable elements, thereby accelerating L2 acquisition relative to L1.

7. The method of claim 6, further comprising accessing a multilingual semantic alignment dataset to identify, for each L2 lexical item, whether a corresponding concept exists in the L1 inventory, and where it exists, grounding the L2 item in the existing concept rather than constructing a new one.

8. The method of claim 6, wherein acquisition of each successive language beyond L2 generally requires a smaller delta than the preceding acquisition, producing an accelerating multilingual dynamic, subject to detection and targeted retraining of negative-transfer cases between particular language pairs.

9. The method of claim 6, wherein phonemes present in L2 but absent from the L1 inventory are identified and subjected to targeted phonemic retraining, while shared phonemes are transferred without retraining.

10. A computer-implemented speech output pipeline configured to produce natural acoustic speech from internalized learned parameters, comprising: a lexical selection stage selecting output word forms from an internalized lexical-semantic space without dictionary lookup; a grapheme-to-phoneme conversion stage producing context-sensitive phoneme sequences from internalized mappings; a coarticulation modeling stage modifying the phoneme sequence for acoustic influence of adjacent phonemes; a prosody generation stage producing prosodic parameters at syllable, word, phrase, sentence, and discourse levels from internalized representations; a register and affect calibration stage; a self-monitoring feedback stage evaluating generated output against internalized acoustic targets and naturalness distributions and applying corrections prior to external output; and a neural vocoder stage rendering acoustic targets, prosodic contours, and affect parameters as a natural speech waveform.

11. The pipeline of claim 10, wherein the coarticulation stage is trained on speech corpora and multimedia dialogue comprising naturalistic connected speech at varying rates, registers, and acoustic conditions.

12. The pipeline of claim 10, wherein prosody generation at all five levels is derived exclusively from internalized representations, without a rule-based prosody system.

13. The pipeline of claim 10, wherein the self-monitoring stage is an internal closed feedback loop executing prior to any external output.

14. A system for co-grounded integration of natural-language understanding with federated data query execution, comprising: a Language Expression Layer performing lexical, morphological, syntactic, and semantic analysis of natural-language input in any supported language; a Federated Query Language (FQL) interface resolving LEL semantic representations to FQL operations against a Universal Federated Connectivity System (UFCS); a UFCS executing FQL queries across federated sources and returning structured results; and a rendering pathway rendering results as natural-language output in the language of the original input, optionally via a speech output pipeline.

15. The system of claim 14, wherein natural-language input in any supported language resolves to an equivalent FQL operation, such that the FQL execution layer is language-agnostic and returns equivalent results regardless of input language.

16. The system of claim 14, wherein the co-grounding comprises a structural correspondence between LEL linguistic representations grounded in entities, relationships, and events and FQL formal representations of entities, relationships, and operations, said correspondence realized as a hybrid of an emergent consequence of the developmental learning sequence and an explicit semantic-alignment mapping layer.

- 17.** The system of claim 14, wherein the LEL acquires all linguistic capabilities as internalized learned parameters without external dictionaries, rule-based modules, or language packs, such that natural-language understanding is fully self-contained within the LEL model.
- 18.** The system of claim 1, further comprising the speech output pipeline of claim 10, driven exclusively by learned parameters internalized within the LEL.
- 19.** The system of claim 1, further comprising the co-grounded integration system of claim 14, wherein the LEL serves as the shared neural foundation for both language acquisition and natural-language query resolution.
- 20.** The system implementing the method of claim 6, wherein multilingual L2+ acquisition extends the language-agnostic query layer of claim 14 to each acquired language without modification of the FQL or UFCS architecture.
- 21.** A computer-implemented pronunciation subsystem comprising: a selector configured to determine, for a supplied word or phrase in a selected language, whether a phonemizer component is available; a phonemizer path that, when the phonemizer is available, obtains a phonemic transcription therefrom; a rule path comprising, for each of a plurality of languages, an ordered set of longest-match grapheme-to-phoneme rules that, when the phonemizer is unavailable, produces an approximate phonemic transcription; and a derivation stage producing a syllable count and a primary-stress position from the transcription; wherein, when neither the phonemizer nor a rule set is available for the selected language, the subsystem returns an explicit null result and does not fabricate a transcription, such that a deterministic transcription is produced offline without any installed engine whenever a rule set exists.
- 22.** The subsystem of claim 21, wherein phonemizer invocations are memoized in a cache and bounded by a timeout such that no invocation blocks an interactive interface, and the syllable count is derived from vowel nuclei of the transcription and the primary-stress position from stress marks of the phonemizer or, for the rule path, from a per-language stress rule.
- 23.** The subsystem of claim 21, further comprising a phrase-level transcription stage that transcribes an entire phrase in a single phonemizer invocation, such that a phonetic transcription presented for display is identical to the transcription from which an audio waveform is subsequently synthesized, thereby preventing divergence between a displayed transcription and spoken output.
- 24.** A computer-implemented speech-output system configured for operation on a machine lacking any previously installed speech facility, comprising a self-contained application bundle that includes, within the application, an open-source speech engine, its runtime-support libraries, and its phoneme data, the system configured to synthesize language-correct audio offline without installation, account, or network service.
- 25.** The system of claim 24, further comprising an engine-selection ladder configured to select, per operating system and per language, among the bundled open-source speech engine, a host operating-system built-in voice service, and a culture-matched operating-system voice where installed, and to report which engine produced each waveform; wherein, when no voice exists for the selected language, the system returns a refusal with a remedial instruction and does not fabricate audio.
- 26.** The system of claim 24, wherein the speech engine is addressed through a role interface and is substitutable with a different speech engine without altering the determinism or provenance guarantees of the system.
- 27.** A computer-implemented translation method comprising: rendering text between a first language and a second language word-by-word through a bundled bilingual dictionary; passing through unchanged, and flagging, each word absent from the dictionary rather than estimating a rendering therefor; and reporting a coverage measure indicating a proportion of words rendered from the dictionary; the method performed deterministically and without a large language model such that the rendering is auditable word-by-word.

28. The method of claim 27, further comprising determining a language of supplied text deterministically by scoring the text against each of a plurality of bundled dictionaries and reporting a ranked estimate with a confidence value.

29. A computer-implemented quality-control method for a pronunciation and speech subsystem, comprising, for each of a plurality of languages and without a large language model: verifying that a phonemic transcription is produced for a set of known words; verifying that a computed syllable count lies within a bounded tolerance of an expected value; verifying that a phrase synthesizes to a valid audio waveform comprising a well-formed container of non-trivial length and decodable duration; and returning a per-language report together with an overall pass score, the method being deterministic and locally executed such that it provides a standing, reproducible verification of the subsystem.

30. The subsystem of claim 21, the system of claim 24, and the methods of claims 27 and 29, wherein each pronunciation, speech, translation, detection, and verification operation is performed without any large language model and each asserted datum is written as a provenance-tracked record carrying an auditable source.

ABSTRACT

A system and method for hierarchical developmental language acquisition. A Sub-Language Priming Layer establishes acoustic, phonemic, prosodic, communicative, and information-theoretic foundations prior to language corpus training. A Language Expression Layer (LEL) acquires all linguistic capabilities — phonetics, phonology, morphology, grammar, syntax, semantics, idioms, and pronunciation — as internalized learned parameters from a structured multilingual corpus including speech and multimedia, without external dictionaries, rule engines, or language packs. A Transition Gate of four measurable thresholds governs progression from structural-acoustic to semantic-communicative learning. A Delta Acquisition Model exploits first-language structural transfer to accelerate multilingual acquisition. A Speech Output Pipeline comprising grapheme-to-phoneme conversion, coarticulation modeling, prosody generation, register calibration, self-monitoring feedback, and neural vocoder rendering produces natural speech from internalized parameters. The LEL is co-grounded with a Federated Query Language and Universal Federated Connectivity System, enabling language-agnostic natural-language input to resolve to structured query execution, with results rendered as natural language or speech in the language of the original query.

COUNSEL FLAG — UFCS/FQL claim scope. Claims 14–20 and Section VI treat FQL/UFCS co-grounding as a core relationship, consistent with the current design in which **UFCS/FQL is integral to the Fact Unit (not optional)**. Note that earlier QueryBook materials (Bible §3.9 / Feature 48) had expressly *removed* UFCS/FQL from the core as a deliberate design-around; re-integrating it as core here is an intentional claim-scope change. Confirm this reversal — and its prior-art/infringement implications — before filing.

COUNSEL FLAG — reduced-to-practice embodiment (Section VII). Section VII ([0019]–[0026]) records a working software embodiment, to strengthen enablement and written description. It does **not** add or amend any claim. The reduction-to-practice assertions are expressly limited to the **deterministic, dependency-free forms** disclosed: the structural-acoustic Phase One and the transition gate ([0019]–[0020]); the deterministic dictionary-and-store semantic grounding of Phase Two ([0021]); the reasoning family — hypothesis, simulation, gated deductive-closure reasoning, and planner ([0023]); the deterministic multilingual delta of Phase Three ([0024]); and the deterministic pronunciation-analysis and operating-system voice forms of Phase Four ([0025]). Expressly **not** asserted as reduced to practice, and described as preferred/prophetic embodiments, are: the trained neural LEL of Section II; the trained-neural forms of Phase-Two grounding and the Delta Acquisition Model; the neural speech-output pipeline of Section V and claims 10–13; and the creativity-parity process, its archive, and its benchmark ([0026]). Counsel to decide whether to (a) retain the prototype disclosure, (b) add dependent claims directed to any deterministic embodiment, and (c) confirm no prophetic embodiment is worded as if actually performed.

COUNSEL FLAG — advanced deterministic embodiment (Section VIII) and new claims 21–30. Section VIII ([0027]–[0031], FIG. 9–10) records further working, deterministic, dependency-free embodiments: the multilingual pronunciation preference ladder (open-source

phonemizer, then per-language rule tables, then a non-fabricated null); the single-call phrase transcription that keeps the displayed transcription consistent with synthesized audio; the self-contained application bundle of a substitutable open-source speech engine with an engine-selection ladder and honest refusal (no fabricated audio); the deterministic translation with coverage and unknown-word passthrough, and the language detector; and the permanent deterministic quality-control battery. **New dependent and independent claims 21–30 are directed to these deterministic embodiments**, as contemplated by flag (b) above. Third-party components are described generically, by role, as substitutable open-source components; counsel to confirm this genericization is adequate for enablement and to review the added claims for scope, antecedent basis, and prior art (expect art in rule-based grapheme-to-phoneme conversion, concatenative/offline TTS packaging, dictionary machine translation, and software test harnesses — the claimed novelty resides in the integrated deterministic, provenance-tracked, LLM-free arrangement and its no-fabrication and substitutability constraints).

DRAFT — NOT FILED — FOR REVIEW BY QUALIFIED PATENT COUNSEL PRIOR TO FILING. Provisional application under 35 U.S.C. § 111(b); not examined. Audit-marked refinements strengthen enablement and are subject to counsel review. A professional prior-art search is required before filing; expect relevant art in self-supervised speech representation learning, curriculum learning, cross-lingual transfer, neural text-to-speech, and developmental language grounding — the claimed novelty resides in the integrated developmental sequence, the threshold-governed transition gate, the delta acquisition model, and the LEL–FQL co-grounding. Inventors, docket number, and filing date to be completed by counsel.